

Add @, \$, and ` to the basic character set

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Contents

- [1 Abstract](#)
- [2 Motivation](#)
- [3 Wording](#)
- [4 References](#)

§ 1 Abstract

WG14, the C Standardization committee, is adopting [N2701] for C23. This will add U+0024 \$ DOLLAR SIGN, U+0040 @ COMMERCIAL AT, and U+0060 ` GRAVE ACCENT to the basic source character set. C++ should adopt the same characters for C++26.

§ 2 Motivation

These characters are available in all encoded character sets in common use and everyone assumes that they are available, using them freely in source text. The primary change would be that these characters become available for syntactic purposes. Although using \$ in identifiers is a common extension, they were not added to the identifier set in C, and this paper does not propose adding them either. Nor were trigraphs added in C for these characters, and this paper does not propose additional trigraphs or digraphs be added.

The translation model for C makes adding these to their basic source character set, the encoded set for source code before translation, much more compelling. These characters being already in the translation character set as single byte characters makes this less important for C++. Nonetheless, it would be useful to make these available for language

purposes as the more conservative C language has agreed there are no functional impediments to their use.

Corentin Jabot discusses the usage in other programming languages extensively in [P2342R0], *For a Few Punctuators More*, q.v.

§ 3 Wording

These changes are relative to [N4901] “Working Draft, Standard for Programming Language C++”

Modify [lex.charset] as follows:

- 2 The basic character set is a subset of the translation character set, consisting of 9699 characters as specified in Table 1.

Modify [tab:lex.charset.basic] with the following additions:

U+0009	CHARACTER TABULATION	
U+000B	LINE TABULATION	
U+000C	FORM FEED	
U+0020	SPACE	
U+000A	LINE FEED	new-line
U+0021	EXCLAMATION MARK	!
U+0022	QUOTATION MARK	"
U+0023	NUMBER SIGN	#
U+0024	DOLLAR SIGN	\$
U+0025	PERCENT SIGN	%
U+0026	AMPERSAND	&
U+0027	APOSTROPHE	'
U+0028	LEFT PARENTHESIS	(
U+0029	RIGHT PARENTHESIS	)
U+002A	ASTERISK	*
U+002B	PLUS SIGN	+
U+002C	COMMA	,
U+002D	HYPHEN-MINUS	-
U+002E	FULL STOP	.
U+002F	SOLIDUS	/
U+0030 .. U+0039	DIGIT ZERO .. NINE	0 1 2 3 4 5 6 7 8 9
U+003A	COLON	:
U+003B	SEMICOLON	;
U+003C	LESS-THAN SIGN	<
U+003D	EQUALS SIGN	=
U+003E	GREATER-THAN SIGN	>
U+003F	QUESTION MARK	?
U+0040	COMMERCIAL AT	@

U+0041 .. U+005A	LATIN CAPITAL LETTER A .. Z	A B C D E F G H I J K L
M		N O P Q R S T U V W X Y
Z		
U+005B	LEFT SQUARE BRACKET	[
U+005C	REVERSE SOLIDUS	\
U+005D	RIGHT SQUARE BRACKET	]
U+005E	CIRCUMFLEX ACCENT	^
U+005F	LOW LINE	_
U+0060	GRAVE ACCENT	`
U+0061 .. U+007A	LATIN SMALL LETTER A .. Z	a b c d e f g h i j k l
m		n o p q r s t u v w x y
Z		
U+007B	LEFT CURLY BRACKET	{
U+007C	VERTICAL LINE	
U+007D	RIGHT CURLY BRACKET	}
U+007E	TILDE	~

§ 4 References

[N2701] Philipp Klaus Krause. @ and \$ in source and execution character set.

<http://www.open-std.org/jtc1/sc22/wg14/www/docs/n2701.htm>

[N4901] Thomas Köppe. 2021-10-22. Working Draft, Standard for Programming Language C++.

<https://wg21.link/n4901>

[P2342R0] Corentin Jabot. 2021-03-25. For a Few Punctuators More.

<https://wg21.link/p2342r0>